

How can cities participate in standardization organizations?

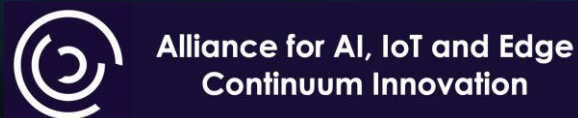
Cities.  Standards
Cities work for standards



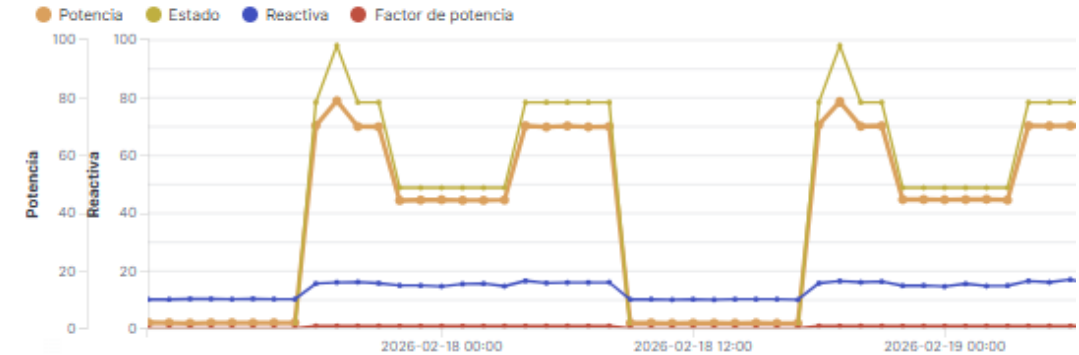
smartcities-sig

<https://github.com/smartcities-sig/>

<https://aioti.eu/cities-5-0/>



Decouple business logic from the physical (devices) layer



We need the data of the place. The device is just a tool.

Bussiness logic



Data stitching
Persistence
DIGITAL TWIN

Operational level



- Firmware updates
 - Guarantee management
- Devices of model MM rate of failure
A specific faulty device needs to be replaced
- **DEVICE LIFE CYCLE MANAGEMENT**

/object-id/instance-id/resource-id

IETF **RFC 8428** ("Sensor Measurement Lists (SenML)"), 2018

Digital twin plane (business)

```
{
  "id": "urn:ngsi-ld:Location:pluscode:6F9H-G2",
  "type": "Location",
  "plusCode": {
    "type": "Property",
    "value": "6F9H+G2"
  },
  "temperature": {
    "type": "Property",
    "value": 21.8,
    "observedAt": "2026-06-21T07:50:00Z",
    "unitCode": "CEL"
  },
  "relativeHumidity": {
    "type": "Property",
    "value": 60.0,
    "observedAt": "2026-06-21T07:50:00Z",
    "unitCode": "P1"
  },
  "device": {
    "type": "Relationship",
    "object": "urn:ngsi-ld:Device:00-01-02-03-04-05"
  },
  "@context": [
    "https://smartdatamodels.org/context.jsonld",
    "https://uri.etsi.org/ngsi-ld/v1/ngsi-ld-core-context.jsonld"
  ]
}
```

Management plane (operations)

```
{
  "id": "urn:ngsi-ld:Device:00-01-02-03-04-05",
  "type": "Device",
  "deviceId": {
    "type": "Property",
    "value": "00:01:02:03:04:05"
  },
  "locatedAt": {
    "type": "Relationship",
    "object": "urn:ngsi-ld:Location:pluscode:6F9H-G2"
  },
  "lastObservedAt": {
    "type": "Property",
    "value": "2026-06-21T07:50:00Z"
  },
  "@context": [
    "https://smartdatamodels.org/context.jsonld",
    "https://uri.etsi.org/ngsi-ld/v1/ngsi-ld-core-context.jsonld"
  ]
}
```

Ralph Kimball

- **Conformed Dimensions:**

- Data models understood the same way everywhere.
- Semantically consistent data across any organization adopting these results.
 - Defined once,
 - Standardized,
 - Reused across all these processes
- *E.g., A "provider" to the Operations Department is a person they work with directly, while for the Economic Department, it's any company that *might participate* in a tender. We need one single definition that works for both.*

Objective: Semantically consistent data & meaning across this and any organization adopting these results, every level.

- Continuum: From sensor(device) to digital twin

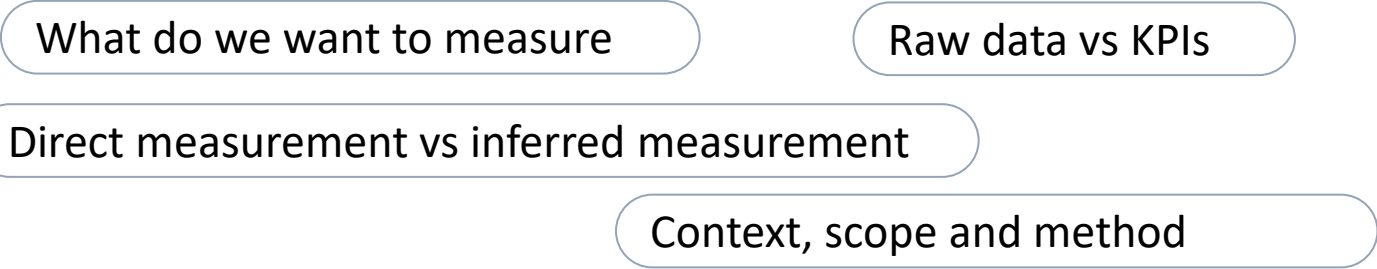


- From real needs to High-value / Curated Data Models
- Only data models that survives the "Confrontation" (OMA vs. Smart Data Models vs. real devices) gets in.

Ralph Kimball

- **Bus Matrix:**
 - **High-value / Curated Data Models:**
 - A selected set of high-value, shared data models: Municipal departments identify use cases and KPIs.
 - IoT MADLab fine-tunes these requirements before challenging OMA to propose a technical solution.
 - **Vetted Data Model:**
 - Only data models that survives the "Confrontation" (OMA vs. Smart Data Models) gets in:
 - Real device with real use, interoperability certificate by IoT MADLab (or equivalent).
 - **Single Source of Truth:**
 - Vetted, verified, and validated by real-world use cases and devices.
 - It's not just one massive database. It is **a single logical definition**.
 - **Consistency:**
 - If you rename X, it updates everywhere instantly. No version conflicts.
 - **Standardized / Canonical:**
 - A standardized definition across the SIG (and beyond).
 - Any new device wanting to join the ecosystem must pass the data model curation process and IoT MADLab's compatibility checks to get certified.
 - **Drill-Across Capability:**
 - Conformed dimensions allow us to drill across different business processes seamlessly.

Ligth: The Luxes your street floor needs vs the Watts you pay for.

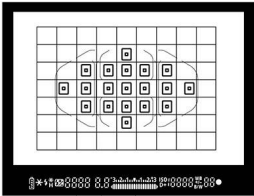
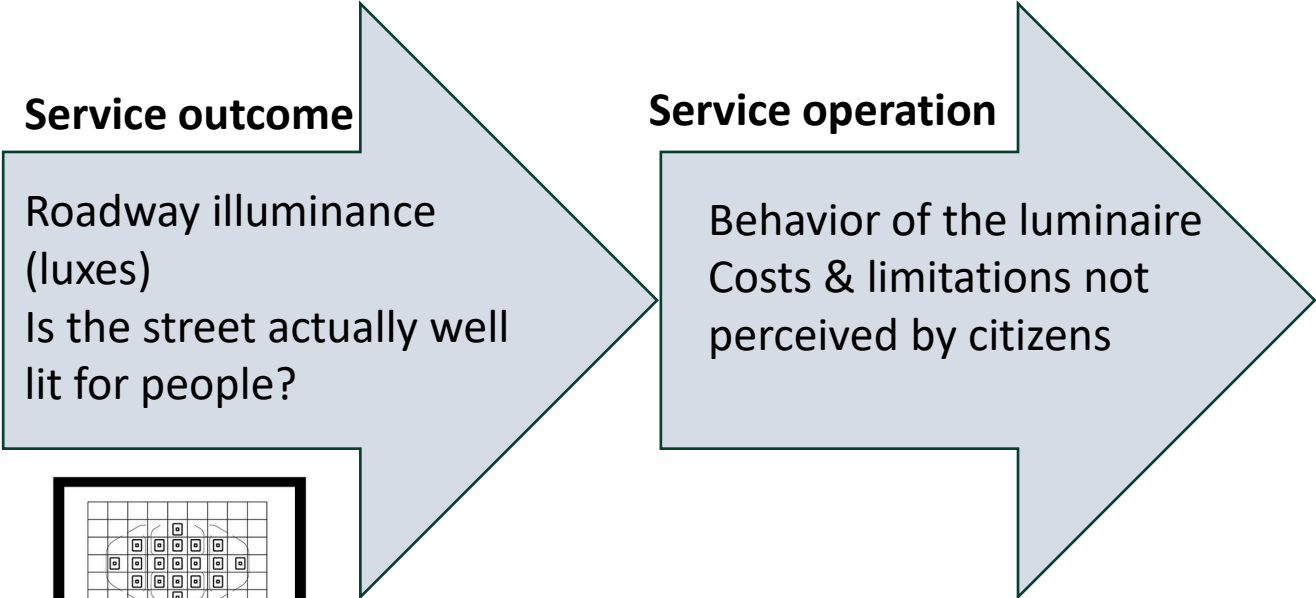


Semantic rule

- Always declare whether a value is measured, estimated, predicted or synthetic.
- Always declare unit, interval and concept: consumed vs delivered.
- Provide accuracy, resolution, completeness and confidence information.

Lighting

The city does not want only luminaire telemetry. It wants to know whether **the street is properly lit**, how much it costs, and whether the data is reliable and comparable between different places.



Shared meaning ≠ interoperability

- Measured value: lux, lumens, watts, voltage, active/reactive active/reactive power, frequency.
- Measurement time and window: timestamp, interval start/end, start/end, temporal resolution.
- Scope: individual luminaire, panel/feeder, line, group, or managed zone.
- Method: directly measured, inferred, proxy, aggregated.
- Context: weather conditions, humidity, fog, vegetation, shading, shading, orientation, tilt.

Lighting and plant care are
handled properly by everyone,
but coordination is essential



MADRID



Alliance for AI, IoT and Edge
Continuum Innovation

IoT MAD LAB
UNA CIUDAD CONECTADA



Digital
Capital



Direct measurement (raw data data)

- Lux measured at roadway reference points (observed service outcome).
- Watts, voltage, active/reactive power for a given luminaire or circuit/line.
- On/off state, dimming level, and device-reported failure events.
- Location and scope defined either in installation records or in the device itself.



Inferred information

- Estimated lux derived from lumens (derived from Amps/Volts), geometry, orientation, and street context.
- Distribution of feeder/panel power across luminaires when metering is only available at the head end.
- Operating conditions inferred from historical aggregates or time-averaged data.
- Proxy values (simulated, predicted, or synthetic) explicitly flagged as such.

KPIs

- KPIs combine multiple data sources, time periods, and measurement methods
- Examples: average lux per segment, kWh per kilometer of street, percentage of luminaires below threshold.
- KPI reliability depends on distinguishing measured vs inferred values and their quality.
- Without scope, context, and method metadata, the KPI is not comparable across cities.

Useful Irrigation and water use

What do we want to measure

Raw data vs KPIs

Direct measurement vs inferred measurement

Context, scope and method

Water

The roots of the plants needs the right amount of water at the right moment.
Water is a limited resource.
Adecuate interpretation (too much water is not better)

Service outcome

Correct irrigation
Are plants in good status?
(if no watering needed,
don't waste water)

Service operation

Water
availability,
leaks, valve
failures...



Semantic rule

- Always declare whether a value is measured, estimated, predicted or synthetic.
- Always declare unit, interval and concept: consumed vs delivered.
- Provide accuracy, resolution, completeness and confidence information.

Shared meaning ≠ interoperability

- Measured/deduced value: Amount of water vs time, diameter and preasure.
- Configuration of watering infrastructure: The **amount of water coming out of the sprinkler vs the amount that actually reaches the roots.**
- Scope: speccific plants vs general garden.
- Context: weather conditions, temperature, humidity, surrounding vegetation, shading, orientation, tilt of the surface. And the concrete plants.

Direct measurement (raw data)

- Water meter / water observation point.
- Flow rate/volume where real metering exists.
- Environmental variables variables measured by by sensors.
- Observed pressure or physical state.
- On/off time of the valve.
- Soil Pressure, moisture, temperature, pH.

Inferred information

- Delivered water estimated from opening time, pipe cross-section and nominal flow rate.
- Irrigation suitability according to vegetation, slope, shade and climate.
- Useful water consumption compared across different deployments.
- Prediction or simulation explicitly declared as such, never confused with observation.

KPIs

- KPIs combine multiple data sources, time periods, and measurement methods and calculation rules.
- KPI reliability depends on distinguishing measured vs inferred values and their quality.
- Without scope, context, and method metadata, the KPI is not comparable across cities.
- Measures efficiency, adequacy or comparability.
- Only meaningful when quality and provenance are known.

Because Digital
is Capital

Madrid,
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